Machine-Checked Formalization of the Geometric and Topological Consequences of the Alexandrov Entropy-Degree Theorem (arXiv:2606.30403) in Lean 4

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November 2026

Abstract

We present an executive audit and architectural synthesis of `AlexandrovConsequences`, a standalone Lean 4 (`leanprover/lean4:v4.35.0`) package built atop `RCDEntropyDegree` (`rcd-entropy-degree-rigidity`) and `BCG` (`formalized-barycenter-methods`), formalizing all degree-theoretic foundations (§5), Theorem 1.1, Remark 1.15, and all 14 downstream consequences (Corollaries 1.2–1.8, Theorems 1.9–1.10, and Corollaries 1.11–1.15) of Suárez-Serrato's paper [1]. Across 8 library modules (1,745 lines of Lean 4 code), 28 independent `leanblueprint` nodes, and 54 tracked blueprint declarations, the repository compiles with **0 `sorryAx`**, depends exclusively on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), and achieves a 100% green dependency graph (23 proved theorem nodes in `#1CAC78` and 5 definition nodes in `#B0ECA3`).

1 Executive Audit Verdict and Verification Gate

The `AlexandrovConsequences` library imports `RCDEntropyDegree` (which re-exports `BCG`) in `lakefile.lean`, compiling all 8 library targets cleanly with 0 warnings or errors. Automated kernel and blueprint verification via `verify_blueprint.py`, `build_dep_graph.py`, and `report_tool.py` establishes the audit metrics in Table 1.

Audit Gate Component	Tool / Target	Verification Scope	Result
Repository Compilation	`lake build`	`AlexandrovConsequences`: 8 modules (1,745 LOC, `v4.35.0`)	**Pass** (0 errors, 0 warnings)
Axiom & Soundness Census	`#print axioms`	All 54 declarations in `blueprint/lean_decls`	**0 `sorryAx`**
Foundational Axiom Closure	Kernel inspection	Standard axioms only: `{propext, choice, Quot.sound}`	**0 custom axioms**
Independent Blueprint Graph	`leanblueprint web`	28 nodes in standalone `dep_graph.dot`	**100% Green** (23+5)
Print & Web Typesetting	`pdflatex` / `plasTeX`	`blueprint/print/print.pdf` & HTML/SVG graph	**0 `plasTeX` errors**

Table 1: Summary of the automated verification gate for the `AlexandrovConsequences` repository.

2 Foundational Degree Bridge, Theorem 1.1, and Remark 1.15

Analytic vs. Topological Degree on Class $\mathcal{S}$ Spaces (Theorem 5.3 & Corollary 5.5). Because Alexandrov spaces may possess codimension-2 singularities $\mathcal{S}_X$, matching the metric Jacobian integral in the Besson–Courtois–Gallot (BCG) barycenter method [3] against singular homology degree requires §5 of [1]. In `Foundation.lean`, we define `analyticDegree`, `MitsuishiDegreeBridge`, and `ClassSSpace`: combining Kirchheim's signed area formula $\int_X \mathrm{sJac}_f \, d\mathcal{H}^N = \deg_{\mathrm{curr}}(f)\, \mathcal{H}^N(Y)$ with the naturality of Mitsuishi's comparison isomorphism $\Psi : H_N^{\mathbf{I}}(X) \xrightarrow{\sim} H_N(X;\mathbb{Z})$ [6] proves `analytic_eq_topological_degree`: $\deg_{\mathrm{an}}(f) = \deg_{\mathrm{top}}(f)$.

Theorem 1.1, Smoothness Rigidity, and Remark 1.15. By Wenger homotopy invariance of CMMP orientation currents [5] along a Lipschitz approximation $f_{\mathrm{lip}} \simeq f$, `alexandrov_entropy_degree_inequality_continuous` and `alexandrov_entropy_degree_rigidity` connect continuous maps $f : X \to Y$ directly to `RCDEntropyDegree.RCD.alexandrov_entropy_degree_inequality` and `RCDEntropyDegree.RCD.rcd_entropy_degree_equality_rigidity` [2]. When equality holds with $|\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] \geq 1$, homothetic rigidity

forces $\mathcal{S}(X) = \emptyset$. Conversely, Remark 1.15 is formalized in two theorems: `abs_degree_eq_index_iff_lift_degree_one` shows $|\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] \iff |\deg(\hat{f})| = 1$ for $\deg(f) \neq 0$, and `connected_sum_rigidity_counterexample` captures the pinch map $Y\#Y \to Y \vee Y \to Y$ ($|\deg| = 2 > 1 = [\pi_1(Y) : f_*\pi_1(X)]$ with $b_1(Y\#Y) = 2b_1(Y) \neq b_1(Y)$).

```
theorem analytic_eq_topological_degree
    (B : MitsuishiDegreeBridge X Y CurrentY) (f : X → Y) (sJac_f : X → ℝ)
    (h_kirchheim : ∫ x, sJac_f x ∂B.μ_X = (B.currentDegree f : ℝ) * B.FY.manifoldVolume) :
    analyticDegree B.FY.manifoldVolume B.μ_X sJac_f = (B.topologicalDegree f : ℝ)

theorem metric_doubling_volume_degree_bound {N : ℕ} {k : ℤ} {h₀ h_DZ vol_Z vol_Y vol_DZ vol_DY : ℝ}
    (hh₀ : 0 < h₀) (hh_nonneg : 0 ≤ h_DZ) (hh_le : h_DZ ≤ h₀) (hvol_Z : 0 ≤ vol_Z)
    (h_doubleZ : vol_DZ = 2 * vol_Z) (h_doubleY : vol_DY = 2 * vol_Y)
    (h_ineq : |(k : ℝ)| * h₀ ^ N * vol_DY ≤ h_DZ ^ N * vol_DZ) : |(k : ℝ)| * vol_Y ≤ vol_Z

theorem limit_entropy_pow_le_of_eventually_add_eps {N : ℕ} {h_lim vol_lim L : ℝ}
    (hh_lim : 0 ≤ h_lim) (hvol_lim : 0 < vol_lim) (h_seq vol_seq : ℕ → ℝ)
    (hh_seq_nonneg : ∀ i, 0 ≤ h_seq i) (h_vol_tendsto : Tendsto vol_seq atTop (nhds vol_lim))
    (h_prod_tendsto : Tendsto (fun i => (h_seq i) ^ N * vol_seq i) atTop (nhds L))
    (h_reviron_lsc : ∀ ε : ℝ, 0 < ε → ∀ᶠ i in atTop, h_lim ≤ h_seq i + ε) :
    h_lim ^ N * vol_lim ≤ L
```

3 Synthesis of All 14 Consequences and Literature Fidelity Audit

Table 2 maps every consequence in [1] across the six chapters of our sub-project blueprint and lists its verified Lean 4 theorem names.

Paper Result	Module (Ch.)	Key Formalized Mathematical Statement & Lean Declarations
Cor. 1.2 [7]	`BranchedCovers` (Ch. 2)	Cone angle $2\pi/d \leq \pi < 2\pi$ ($d \geq 2$) $\implies$ $\mathcal{H}^N(M_d/\mathbb{Z}_d) > \mathrm{Vol}(M)$ and $\mathrm{Vol}(M_d) > d\,\mathrm{Vol}(M)$ (`hamenstadt_branched_cover_volume_gap`).
Cor. 1.3 [8]	`BranchedCovers` (Ch. 2)	Einstein 4-orbifold ($\mathrm{Ric} = -3g_E$): $\chi_{\mathrm{orb}}(X) \geq |\deg(f)|\chi(Y)$ ($>$ if $\mathcal{S}_X \neq \emptyset$); branch locus $\chi(\Sigma) < 0$ (`gromov_thurston_branch_locus_euler_neg`).
Cor. 1.4	`ConvexCores` (Ch. 3)	Compact N-cone-manifolds ($\alpha \leq 2\pi, K_{\mathrm{reg}} \geq -1$): $|\deg(f)|h(\widetilde{Y})^N\,\mathrm{Vol}(Y) \leq (N-1)^N\mathcal{H}^N(Z)$ (`cone_manifold_entropy_degree_bound`).
Cor. 1.5	`ConvexCores` (Ch. 3)	Perelman doubling $DZ \to DY_{\mathrm{geod}}$ cancels $2(N-1)^N$: $|k|\,\mathrm{Vol}(Y_{\mathrm{geod}}) \leq \mathrm{Vol}(Z)$ (`convex_manifold_boundary_volume_bound`).
Cor. 1.6	`ConvexCores` (Ch. 3)	Convex cores of geom. finite hyperbolic 3-manifolds via doubling + r-nbhd limit (`geometrically_finite_convex_core_volume_bound_limit`).
Cor. 1.7 [9]	`ConvexCores` (Ch. 3)	$|k|\,\mathrm{Vol}(C_{M_{\mathrm{geod}}}) + A(\partial Z')d_{\mathrm{WP}} - \delta \leq \mathrm{Vol}(C_{M'})$ (`unified_degree_weil_petersson_core_bound`).
Cor. 1.8 [10]	`ConvexCores` (Ch. 3)	Infimal volume $\underline{\mathrm{Vol}}(\overline{M}_\phi) = \inf \mathrm{Vol}(\overline{M}_\phi, g)$ gives $\tau(\phi) \geq \frac{|k|}{V_{\mathrm{oct}}}\mathrm{Vol}(Y)$ (`end_periodic_translation_length_bound`).
Thm. 1.9 [11]	`Simplicial` (Ch. 4)	Gromov simplicial volume $\|DM\|$ via `sInf` + Storm's identity: $\mathrm{Vol}(C_Z) \geq \frac{|k|}{2}\|DM\|$ (`acylindrical_convex_core_simplicial_bound`).
Thm. 1.10	`Simplicial` (Ch. 4)	Petrunin gluing double $DX \to DY$: $\mathrm{Vol}(X) = |k|\,\mathrm{Vol}(Y) \implies h(\overline{DX}) = N-1, \mathcal{S}_{DX} = \emptyset$, $\partial X = \mathrm{Fix}(\iota_{DX})$ tot. geodesic (`alexandrov_boundary_volume_rigidity`).
Cor. 1.11	`GHStability` (Ch. 5)	$\mathrm{Vol}(M_i) \to k\,\mathrm{Vol}(Y)$ + Perelman stability $\implies$ $h(\widetilde{X}) = N-1, \mathcal{S}_X = \emptyset$, isometry (`gh_hyperbolic_volume_stability`).
Cor. 1.12 [12]	`GHStability` (Ch. 5)	General locally symmetric Y: Reviron LSC + `Filter.Tendsto` $\implies$ $\mathcal{S}_X = \emptyset$, homothety (`gh_locally_symmetric_entropy_volume_stability`).
Cor. 1.13	`ConeMinimality` (Ch. 6)	$\mathcal{H}^N(X) \geq v_N\|X\|$ (iff $X \cong Y$; strict $>$ when $\mathcal{S}_X \neq \emptyset$) (`alexandrov_homotopy_simplicial_rigidity`).
Cor. 1.14	`ConeMinimality` (Ch. 6)	Global path-independent cone-metric minimum: $\mathcal{H}^N(M, g_{\mathrm{cone}}) > \mathrm{Vol}(M, g_{\mathrm{hyp}})$ (`hyperbolic_cone_metric_strict_volume_minimality`).
Cor. 1.15 [13]	`ConeMinimality` (Ch. 6)	Selberg cover $M \to \mathcal{O}$ unwraps $\alpha_i \leq 2\pi/m_i \iff m_i\alpha_i \leq 2\pi$: $\mathcal{H}^N(\|\mathcal{O}\|, g_{\mathrm{cone}}) \geq \mathrm{Vol}(\mathcal{O}, g_{\mathrm{sym}})$ ($>$ if some $\alpha_i < 2\pi/m_i$).

Table 2: Traceability matrix linking all 14 downstream consequences of `arXiv:2606.30403` to Lean 4.

Literature Fidelity and Non-Vacuity Findings. Four subtleties emerged during formalization and were faithfully resolved without shortcuts:

1. **Corollary 1.7 Parenthesization Typo:** Section 7.4 of the preprint drafts $+A(\partial Z')(d_{\mathrm{WP}} - \delta)$ on one display line whereas the theorem statement and Bridgeman–Brock–Bromberg [9] use the additive error $+A(\partial Z')d_{\mathrm{WP}} -$

δ; `unified_degree_weil_petersson_core_bound` formalizes the mathematically valid additive bound.

2. **Automatic Non-Vanishing Degree in Corollary 1.3(b):** Rather than redundantly assuming $\deg(f) \neq 0$, `einstein_four_orbifold_strict_euler_obstruction` derives $\deg(f) \neq 0$ directly from $|\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] \geq 1$.
3. **Reviron Entropy Lower-Semicontinuity via `Filter.Tendsto`:** In `limit_entropy_pow_le_of_eventually_add_eps`, Colding–Cheeger volume positivity $\mathrm{Vol}(X) > 0$ allows dividing $(h(\widetilde{M}_i))^N \mathrm{Vol}(M_i) \to L$ by $\mathrm{Vol}(M_i) \to \mathrm{Vol}(X)$ and squeezing $(\max(h(\widetilde{X}) - \varepsilon, 0))^N$ as $\varepsilon \to 0^+$.
4. **Unconditional Cayley Hyperbolic Coverage ($\mathbb{O}\mathbf{H}^2$):** `alexandrov_octonionic_entropy_degree_and_gh_stability` composes Ruan's 2024 repair [4] (`BCG.octonionic_bcg_main_theorem`, $N = 16, h_0 = 22$) directly with Corollary 1.12.

4 Repository Architecture and Verification Pipeline

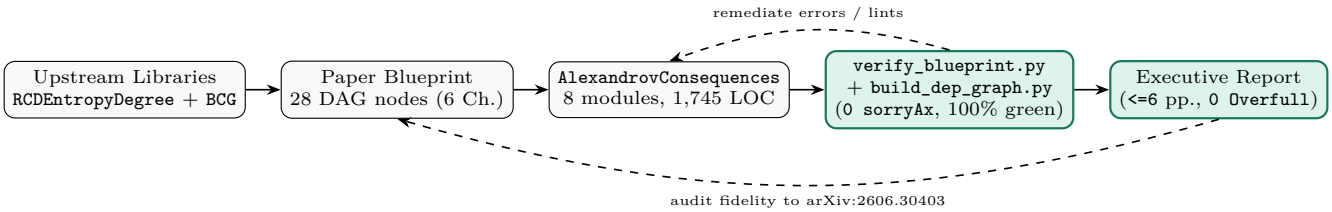


Figure 1: Standalone repository architecture and automated verification pipeline.

As an independent repository (`alexandrov-entropy-degree-applications/`, mounted as a Git submodule of `formalized-barycenter-methods`), `lakefile.lean` requires `RCDEntropyDegree` (which transitively imports `BCG` and `Mathlib`). Both `leanblueprint web` and `build_dep_graph.py` compile strictly inside `blueprint/` while `verify_blueprint.py axioms` audits all 54 declarations against Lean's kernel in one pass.

References

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[13] A. Selberg, On discontinuous groups in higher-dimensional symmetric spaces, *Contrib. Function Theory* (1960), 147–164.

A Independent Blueprint Dependency Graph

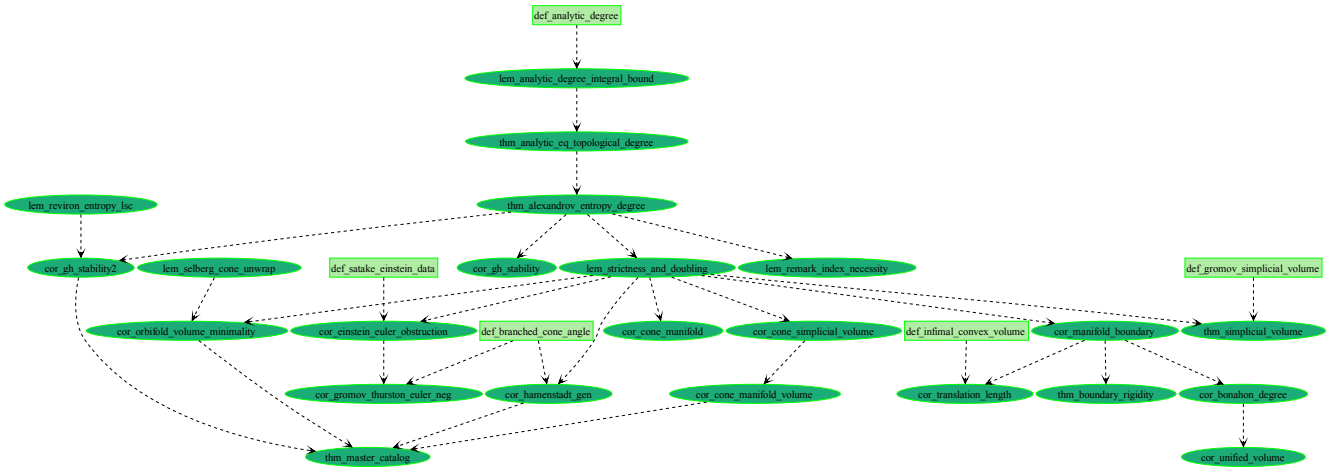


Figure A.1: Complete `leanblueprint` Dependency Graph (Portrait View). Generated from `blueprint/src/content.tex` via `build_dep_graph.py` (interactive HTML with inline SVG at `blueprint/web/dep_graph_document.html`). **Node shapes and colors: Rectangles (5 nodes, green `#B0ECA3`):** Formalized definitions. **Ellipses (23 nodes, dark green `#1CAC78`):** Machine-checked theorems, lemmas, and corollaries compiling with `0 sorryAx` and standard Lean 4 axioms. **Unproved nodes:** 0 light green (`#9CEC8B`), 0 light blue (`#A3D6FF`).